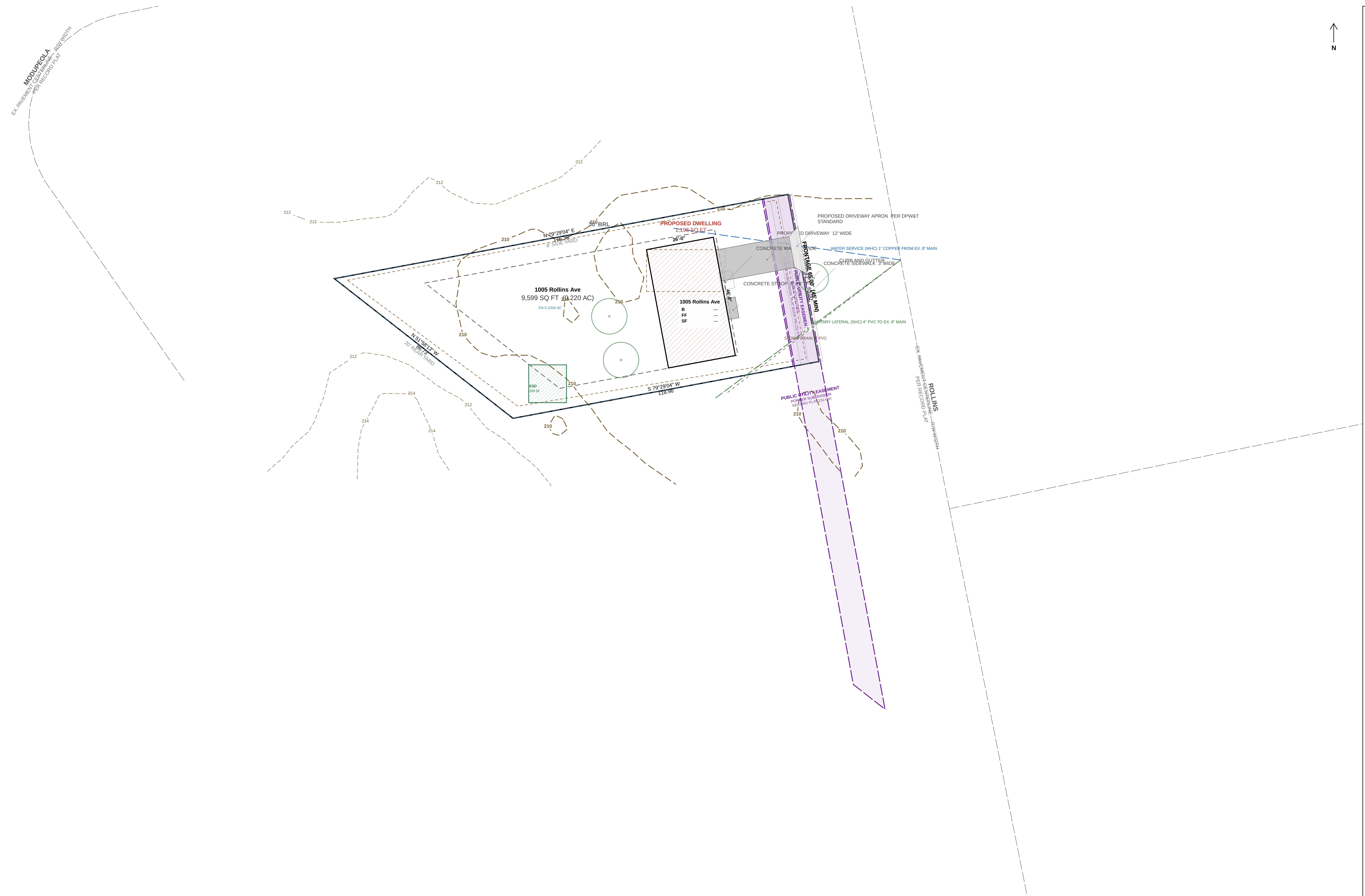
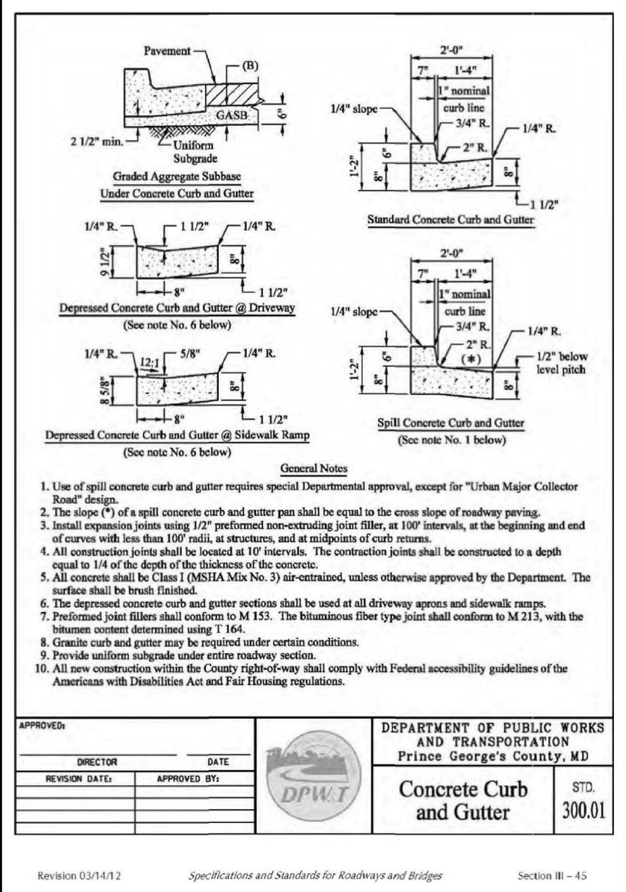


CONCRETE CURB AND GUTTER — PGC DPW&T STD. 300.01


Pavement
(B)
GASB
6"
2 1/2" min.
Uniform Subgrade
2'-0"
7"
1'-4"
1" nominal curb line
1/4" slope
3/4" R.
1/4" R.
2" R.
1'-2"
6"
8"
1 1/2"
1/4" R.
1 1/2"
9 1/2"
8"
12:1
5/8"
8 5/8"
1/2" below level pitch
(*)

Graded Aggregate Subbase Under Concrete Curb and Gutter

Standard Concrete Curb and Gutter

Depressed Concrete Curb and Gutter @ Driveway
(See note No. 6 below)

Spill Concrete Curb and Gutter
(See note No. 1 below)

Depressed Concrete Curb and Gutter @ Sidewalk Ramp
(See note No. 6 below)

General Notes

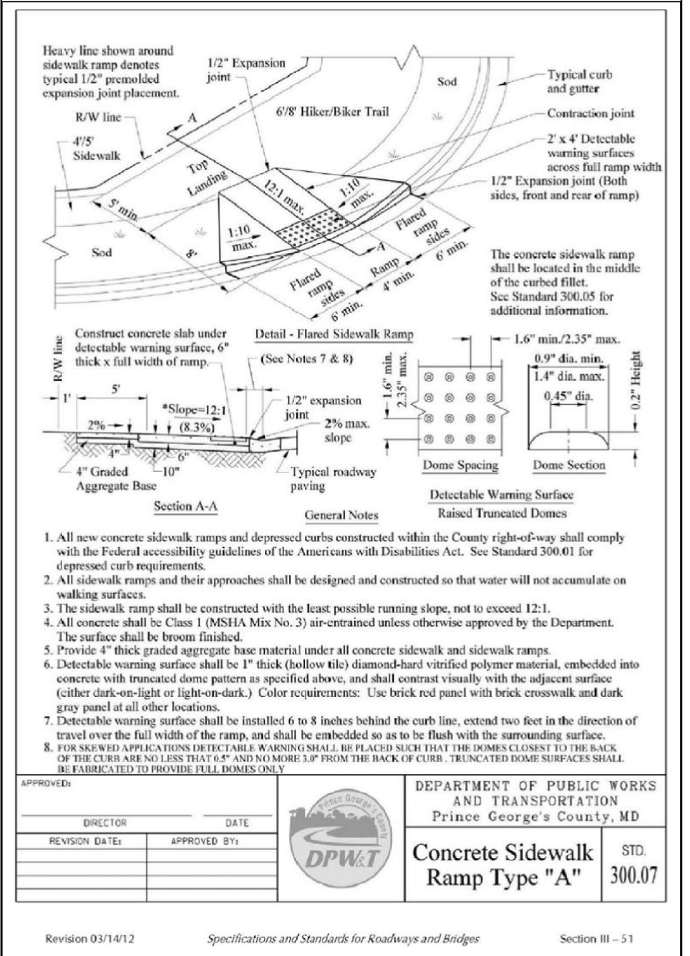
1. Use of spill concrete curb and gutter requires special Departmental approval, except for "Urban Major Collector Road" design.
2. The slope (*) of a spill concrete curb and gutter pan shall be equal to the cross slope of roadway paving.
3. Install expansion joints using 1/2" preformed non-extruding joint filler, at 100' intervals, at the beginning and end of curves with less than 100' radii, at structures, and at midpoints of curb returns.
4. All construction joints shall be located at 10' intervals. The contraction joints shall be constructed to a depth equal to 1/4 of the depth of the thickness of the concrete.
5. All concrete shall be Class I (MSHA Mix No. 3) air-entrained, unless otherwise approved by the Department. The surface shall be brush finished.
6. The depressed concrete curb and gutter sections shall be used at all driveway aprons and sidewalk ramps.
7. Preformed joint fillers shall conform to M 153. The bituminous fiber type joint shall conform to M 213, with the bitumen content determined using T 164.
8. Granite curb and gutter may be required under certain conditions.
9. Provide uniform subgrade under entire roadway section.
10. All new construction within the County right-of-way shall comply with Federal accessibility guidelines of the Americans with Disabilities Act and Fair Housing regulations.

APPROVED:		DEPARTMENT OF PUBLIC WORKS AND TRANSPORTATION Prince George's County, MD	
DIRECTOR	DATE	Concrete Curb and Gutter	STD. 300.01
REVISION DATE:	APPROVED BY:		

Revision 03/14/12 *Specifications and Standards for Roadways and Bridges* Section III – 45

REPRODUCED FROM THE PRINCE GEORGE'S COUNTY DPW&T STANDARD. THE STANDARD GOVERNS.

CONCRETE SIDEWALK RAMP TYPE A — PGC DPW&T STD. 300.07


1/2" Expansion joint
6'/8' Hiker/Biker Trail
Sod
Typical curb and gutter
Contraction joint
2' x 4' Detectable warning surfaces across full ramp width
1/2" Expansion joint (Both sides, front and rear of ramp)
R/W line
A
4'/5' Sidewalk
Top Landing
12:1 max.
1:10 max.
5' min.
1:10 max.
Flared ramp sides
6' min.
Ramp 4' min.
Flared ramp sides
6' min.
Sod
8'
A
R/W line
1'
5'
2%
*Slope=12:1 (8.3%)
1/2" expansion joint
2% max. slope
4"
6"
10"
4" Graded Aggregate Base
Typical roadway paving
1.6" min./2.35" max.
0.9" dia. min.
1.4" dia. max.
0.45" dia.
1.6" min.
2.35" max.
0.2" Height

Heavy line shown around sidewalk ramp denotes typical 1/2" premolded expansion joint placement.

The concrete sidewalk ramp shall be located in the middle of the curbed fillet. See Standard 300.05 for additional information.

Detail - Flared Sidewalk Ramp

Construct concrete slab under detectable warning surface, 6" thick x full width of ramp.

(See Notes 7 & 8)

Section A-A

Dome Spacing

Dome Section

Detectable Warning Surface
Raised Truncated Domes

General Notes

1. All new concrete sidewalk ramps and depressed curbs constructed within the County right-of-way shall comply with the Federal accessibility guidelines of the Americans with Disabilities Act. See Standard 300.01 for depressed curb requirements.
2. All sidewalk ramps and their approaches shall be designed and constructed so that water will not accumulate on walking surfaces.
3. The sidewalk ramp shall be constructed with the least possible running slope, not to exceed 12:1.
4. All concrete shall be Class 1 (MSHA Mix No. 3) air-entrained unless otherwise approved by the Department. The surface shall be broom finished.
5. Provide 4" thick graded aggregate base material under all concrete sidewalk and sidewalk ramps.
6. Detectable warning surface shall be 1" thick (hollow tile) diamond-hard vitrified polymer material, embedded into concrete with truncated dome pattern as specified above, and shall contrast visually with the adjacent surface (either dark-on-light or light-on-dark.) Color requirements: Use brick red panel with brick crosswalk and dark gray panel at all other locations.
7. Detectable warning surface shall be installed 6 to 8 inches behind the curb line, extend two feet in the direction of travel over the full width of the ramp, and shall be embedded so as to be flush with the surrounding surface.
8. FOR SKEWED APPLICATIONS DETECTABLE WARNING SHALL BE PLACED SUCH THAT THE DOMES CLOSEST TO THE BACK OF THE CURB ARE NO LESS THAT 0.5" AND NO MORE 3.0" FROM THE BACK OF CURB . TRUNCATED DOME SURFACES SHALL BE FABRICATED TO PROVIDE FULL DOMES ONLY

APPROVED:		DEPARTMENT OF PUBLIC WORKS AND TRANSPORTATION Prince George's County, MD	
DIRECTOR	DATE	Concrete Sidewalk Ramp Type "A"	STD. 300.07
REVISION DATE:	APPROVED BY:		

Revision 03/14/12 *Specifications and Standards for Roadways and Bridges* Section III – 51

REPRODUCED FROM THE PRINCE GEORGE'S COUNTY DPW&T STANDARD. THE STANDARD GOVERNS.

GRAPHIC SCALE — 1" = 20'

0 20 40 60 80 FEET

KEALEE

design, build, deliver

PROJECT
1005 Rollins Ave

ADDRESS
1005 Rollins Ave

JURISDICTION
prince georges md

ZONE
RSF-65

SHEET
C-001 — Site Development Plan — Existing, Proposed, Grading, Paving and Landscape

SCALE
1" = 20'

STATUS
PRELIMINARY

SHEET NO.
1 OF 1

COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

PLAN TYPE AND PREPARATION	
PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-001 — Site Development Plan — Existing, Proposed, Grading, Paving and Landscape
DISCIPLINE	Maryland Professional Engineer / Surveyor (see divided responsibility)
DRAWN BY	KEALEE SITE-PLAN ENGINE (AUTOMATED)
DESIGNED BY	— (TO BE COMPLETED BY THE DESIGN PROFESSIONAL)
CHECKED BY	— (TO BE COMPLETED BY THE DESIGN PROFESSIONAL)

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA		
ZONE	RSF-65	
TAX / PARCEL ID	1005 Rollins Ave	
GROSS LOT AREA	9,599 SF	0.220 AC
LOT FRONTAGE	45' MIN	65.00'
SETBACKS	**REQUIRED**	**PROVIDED**
FRONT YARD	30'	30'
SIDE YARD	8'	8'
REAR YARD	20'	20'
COVERAGE	**MAX**	**PROPOSED**
LOT COVERAGE	35%	12.5%
BUILDING FOOTPRINT	—	1,196 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	9,599 SF
2. Area of dwelling	1,196 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net tract area	9,599 SF
6. TOTAL AREA DISTURBED	6,564 SF (AT LEAST)
7. SWM REQUIRED — Sec. 32-174(a)(3)	YES — DISTURBANCE EXCEEDS 5,000 SF

BUILDING DATA

	G	B	FF	SF	HT	STY
1005 Rollins Ave	—	—	—	—	20'	2

G garage slab · B basement · FF finished floor · SF subfloor · HT height · STY storeys. A DASH IS NOT ZERO: the elevation has not been established and must be set from a field-run topographic survey before construction.

DRAINAGE AREA COMPUTATIONS

CATCHMENT	**SF**	**ACRES**	**C**
Total — 17.0% impervious	9,599	0.220	
PRE-DEVELOPMENT	**SF**		**C**
Pervious — lawn / meadow	9,599		0.25
Composite C			**0.250**
POST-DEVELOPMENT	**SF**		**C**
Proposed roof	1,196		0.95
Driveway and on-lot paving	438		0.95
Pervious — lawn	7,965		0.25
Composite C			**0.369**
Flow path			204 ft
Flow path slope			2.0 %
Time of concentration (Kirpich)			5.0 min
Rainfall intensity, 10-yr			6.71 in/hr
Peak discharge PRE, Q = CiA			0.37 cfs
Peak discharge POST, Q = CiA			0.55 cfs
INCREASE TO BE MANAGED			**0.18 cfs**
Rv = 0.05 + 0.009 I			0.2032
Water quality volume WQv			**163 cf**

Time of concentration by Kirpich over a 204 ft flow path at 2.0%, taken from the 2 ft contours across the catchment. A 5-minute floor applies.

Rainfall intensity 6.71 in/hr for the 10-year storm at a 5.0 minute duration. NOAA Atlas 14 Vol. 2 Ver. 3, PFDS point estimate, 38.8752 N 76.9019 W.

Water quality volume on a 1.0 in rainfall. Maryland Stormwater Design Manual, Vol. I & II — WQv on a 1.0 in rainfall.

Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.

Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed area.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting	**1 DAY**
2. Obtain necessary permits	**2 DAYS**
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation	**1 DAY**
4. Install stabilized construction entrance and perimeter sediment control	**2 DAYS**
5. Rough grade the lot	**5 DAYS**
6. Construct dwelling	**6 MONTHS**
7. Install water, sewer and storm services and the driveway	**10 DAYS**
8. Fine grade and stabilize all disturbed areas	**5 DAYS**
9. Remove sediment control once the inspector gives written permission	**1 DAY**

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION. FIELD-VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. ALL EXISTING AND PROPOSED UTILITIES SHOWN PER PGC CODE SEC. 32-106.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS THE RECORDED PLAT OF SUBDIVISION — PORTER SUBDIVISION, PLAT BOOK PM 231 P.50 — TRANSCRIBED AND CHECKED FOR CLOSURE. THE PLAT IS THE BOUNDARY SURVEY. NO FIELD TOPOGRAPHY IS INCLUDED; EXISTING GRADE IS COUNTY LIDAR CONTOUR MAPPING.
6. CONNECT TO EXISTING PAVEMENT, CURB AND GUTTER, DRIVEWAY AND SIDEWALK IN LINE AND GRADE.

LEGEND

━━━	PROPERTY BOUNDARY
- - - -	BRL — BUILDING RESTRICTION LINE
━━━	PROPOSED STRUCTURE
━━━	EXISTING CONTOUR — 2 FT INTERVAL, NAVD88
- - - -	STREET CENTRELINE
━━━	PROPOSED PAVEMENT — DRIVEWAY / WALK
- - - -	EASEMENT

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE ____________________ DATE ____________

LICENSE NO. ____________________ EXPIRATION DATE ____________

SEAL

REVISIONS
— none —